

Attachment 2

Wastewater Pretreatment Plant (WWPP) Chemical Treatment Process Wiping Solution Wastewater Pretreatment Plant (WSWPP)

PROCESS DESCRIPTION: The Wiping Solution Wastewater Pretreatment Plant (WSWPP) is designed to remove the ink solids, minor amounts of oil and grease in the Bureau's process wastewater discharge, and control the pH limits. The influent to the WSWPP consists of a waste stream that is composed of approximately (1.0%) aqueous sodium hydroxide, emulsified printing inks, approximately (0.5%) sodium-sulfated castor oil, small quantities of solvents added during press cleaning operations, hydraulic oils from press leaks, and other similar materials. In the WSWPP the wastewater is either pumped or drained from the presses into tank T-1 which serves as a surge tank. The wastewater and ink sludge is then pumped into tank T-3. Bentonite slurry is added to the waste stream prior to the pump so that it can be mixed in line by the pump. The bentonite slurry is added to the waste stream to prevent the sludge from being "tacky" after the acid is added. When too much Bentonite is added, the result is soft filter cakes. When too little Bentonite slurry is added, the ink precipitates and becomes non-pumpable. In tank T-3, 93% sulfuric acid is used to adjust the pH from approximately 13 to 8.5 s.u. In the next tank T-4, the pH is finely adjusted from 8.5 to the final of 8.0. Tanks T-5 and T-6 are used as spare process tanks and are connected to T-4 and T-7. In the next tank T-7, a flocculating polymer is added. The wastewater then flows to one of the two gravity settling units (GSU) or clarifiers. The supernatant water from the GSU then flows to tank T-8 and T-9 and is then pumped through 25µm bag filters to polish the effluent before it is discharged to the sanitary sewer. The sludge is pumped from the bottom of the GSU to a sludge storage and dewatering tank T-10. The sludge in T-10 is pumped to sludge filter presses where the sludge is concentrated into sludge cakes for disposal. The water discharged from the presses flows to tank T-17 (below grade) and is then pumped back to T-7 for treatment. Spills from the system are collected by the floor sumps and returned to the system for pre-treatment. For any major spill within the plant, a truck well is designed to collect and contain the spill.

T-11 was added to collect wastewater from the press bases which contain substantial amounts of oil/water and collect wastewater from the Wiping Solution Production Plant (WSPP). T-11 acts as an oil/water separator. The wastewater exits T-11, leaving the oil in the tank, and enters the equalization tank T-1. The oil is periodically pumped out of T-11. The wastes are disposed of in accordance with EPA regulations.

PROCESS OPERATION: The Contractor shall operate the WSWPP with prudent engineering and operating practices. Operating the WSWPP includes, ***but is not limited to***, the following tasks:

- (1) The Contractor shall take readings every two (2) hours of the pumps data, flows, tank levels, data set points and operating conditions from the computer, and other set points throughout the WSWPP. It should be noted that in the future this may be at least partially performed by SCADA software.
- (2) The Contractor shall perform an inspection of the WSWPP every two (2) hours minimally or more often as needed for proper operation.

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- (3) The Contractor shall perform a jar test every two (2) hours with samples of T-7 and T-10 to determine if the sludge is "tacky" and to determine the settling rate and set points (operating parameters) of the WSWPP.
- (4) The Contractor shall start, stop, empty filter presses, prepare the pallets, boxes, liners, fill each box to between >2,350 pounds and $\leq 2,400$ pounds, band, properly label, and enter the data into the computer logging system which is furnished by the BEP, for the performance of this contract.
- (5) The Contractor shall change/clean filter press filter cloths as needed when they become blinded or physically damaged.
- (6) The Contractor shall mix the system materials as needed for the optimal operation of the system.
- (7) The Contractor shall check pH set points, loop set points, and local set points; and calibrate pH meters weekly at a minimum or more often, as needed.
- (8) The Contractor shall respond to all alarms, correcting the conditions to assure proper operation of the WSWPP.
- (9) The Contractor shall clean and inspect the tanks at least bi-annually (twice per year) as well as clean the tanks if there is a blockage (build-up) of sludge in any tank as necessary during the operation of the WSWPP.
- (10) The Contractor shall order the materials/supplies as needed to assure continuous operation of the WSWPP.

